



Problem Challenge 2

We'll cover the following ^

- Path with Maximum Sum (hard)
- Try it yourself

Path with Maximum Sum (hard)

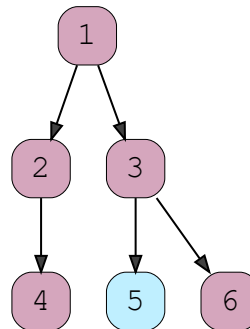
Find the path with the maximum sum in a given binary tree. Write a function that returns the maximum sum.

A path can be defined as a **sequence of nodes between any two nodes** and doesn't necessarily pass through the root. The path must contain at least one node.

Example 1:

Output: 16

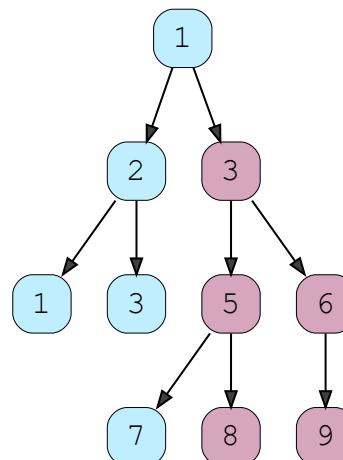
Explanation: The path with maximum sum is: [4, 2, 1, 3, 6]



Example 2:

Output: 31

Explanation: The path with maximum sum is: [8, 5, 3, 6, 9]



Tt



Try it yourself

Try solving this question here:



Java

Python3

JS

C++

```
1 class TreeNode {
2     int val;
3     TreeNode left;
4     TreeNode right;
5
6     TreeNode(int x) {
7         val = x;
8     }
9 };
10
11 class MaximumPathSum {
12
13     public static int findMaximumPathSum(TreeNode root) {
14         // TODO: Write your code here
15         return -1;
16     }
17
18     public static void main(String[] args) {
19         TreeNode root = new TreeNode(1);
20         root.left = new TreeNode(2);
21         root.right = new TreeNode(3);
22         System.out.println("Maximum Path Sum: " + MaximumPathSum.findMaximumPathSum(root));
23
24         root.left.left = new TreeNode(1);
25         root.left.right = new TreeNode(3);
26         root.right.left = new TreeNode(5);
27         root.right.right = new TreeNode(6);
28         root.right.left.left = new TreeNode(7);
29         root.right.left.right = new TreeNode(8);
30         root.right.right.left = new TreeNode(9);
31         System.out.println("Maximum Path Sum: " + MaximumPathSum.findMaximumPathSum(root));
32
33         root = new TreeNode(-1);
34         root.left = new TreeNode(-3);
35         System.out.println("Maximum Path Sum: " + MaximumPathSum.findMaximumPathSum(root));
36     }
37 }
38
```

Run

Save

Reset

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